

SERAKHS, Turkmenistan

WMO: 389740

Lat: 36.533N

Lon: 61.217E

Elev: 275

StdP: 98.06

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.3	-3.8	-12.5	1.3	-0.1	-9.8	1.7	4.7	9.0	15.1	7.5	13.7	1.6	320	0.426

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.5	41.1	20.2	39.7	19.6	38.4	19.3	21.8	36.9	21.1	36.4	20.4	35.5	2.9	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.8	12.4	27.6	15.6	11.5	27.0	14.6	10.7	26.8	65.1	37.2	62.2	36.4	59.8	35.7	26.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.2	6.5	5.4	-9.5	43.7	4.5	1.3	-12.7	44.7	-15.4	45.5	-17.9	46.2	-21.1	47.2		
(5)			-10.2	23.5	4.3	1.1	-13.4	24.3	-15.9	25.0	-18.3	25.7	-21.5	26.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	18.7	6.1	7.6	12.9	18.9	25.2	29.7	31.3	29.7	24.8	18.7	11.8	7.2		
	DBStd	10.05	5.77	5.71	5.32	5.04	3.63	2.64	2.35	2.55	3.32	4.61	5.19	5.06		
	HDD10.0	437	147	101	29	2	0	0	0	0	0	2	38	117		
	HDD18.3	1523	381	301	182	53	3	0	0	0	1	52	205	346		
	CDD10.0	3624	26	34	121	271	472	591	661	611	444	271	92	31		
	CDD18.3	1668	1	1	14	71	217	341	403	353	196	63	8	1		
	CDH23.3	22660	3	12	167	746	2582	4959	6199	4968	2235	685	96	8		
	CDH26.7	13344	0	2	51	319	1364	3053	4024	3089	1144	268	29	1		
(14) Wind	WSAvg	2.2	2.1	2.3	2.3	2.3	2.4	2.6	2.6	2.4	2.0	1.9	2.0	2.1		
(15) Precipitation	PrecAvg	203	31	33	45	38	13	0	0	0	1	4	12	24		
	PrecMax	312	92	80	102	108	49	10	1	5	19	40	54	61		
	PrecMin	127	4	4	8	1	0	0	0	0	0	0	0	0		
	PrecStd	45	21	20	23	26	15	2	0	1	3	8	13	18		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.2	25.0	31.3	36.2	40.0	42.3	43.0	42.1	38.9	35.4	30.5	24.7		
		MCWB	11.2	13.4	15.8	17.9	19.3	20.5	20.9	20.2	18.6	17.2	15.1	12.1		
	2%	DB	20.2	21.7	27.6	33.0	37.4	40.2	41.3	40.0	36.6	32.2	25.8	20.8		
		MCWB	10.1	11.2	14.4	17.6	18.9	19.7	20.5	19.4	17.9	16.5	13.2	10.3		
	5%	DB	17.3	18.9	24.6	30.5	35.3	38.7	39.9	38.5	34.7	29.7	22.9	18.1		
		MCWB	8.9	10.2	13.6	16.6	18.4	19.2	20.0	19.1	17.4	15.4	12.4	9.2		
	10%	DB	14.3	16.2	21.7	27.7	33.4	37.3	38.6	37.1	32.8	27.1	20.1	15.3		
		MCWB	7.8	9.4	12.8	15.7	17.8	18.9	19.7	18.6	16.9	14.7	11.4	8.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.1	13.9	17.2	20.2	21.6	22.5	23.4	22.3	20.3	19.0	16.0	13.0		
		MCDB	20.8	22.4	26.8	30.1	35.2	37.9	37.9	37.0	34.1	31.9	26.3	22.6		
	2%	WB	10.5	12.2	15.4	18.6	20.1	21.1	22.1	21.2	19.2	17.6	14.4	11.2		
		MCDB	18.3	19.7	24.3	29.3	33.1	37.2	37.2	36.1	33.0	29.5	23.1	19.1		
	5%	WB	9.3	10.9	14.3	17.4	19.3	20.4	21.3	20.3	18.3	16.4	13.3	9.9		
		MCDB	16.2	17.2	22.6	27.7	31.8	36.0	36.8	35.5	32.3	27.5	21.0	16.7		
	10%	WB	8.0	9.7	13.2	16.4	18.6	19.7	20.6	19.4	17.5	15.4	12.1	8.7		
		MCDB	13.7	15.4	20.6	25.9	30.6	34.8	36.3	34.9	31.2	25.5	19.1	14.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.0	9.7	10.7	11.8	13.2	13.5	13.5	14.0	14.1	13.3	11.0	9.2		
		MCDBR	13.5	14.3	15.8	16.2	16.1	15.7	15.1	15.4	16.2	16.6	15.2	13.3		
	5% WB	MCWBR	6.3	6.7	6.3	5.9	5.4	5.3	5.0	5.5	6.3	7.3	6.8	6.3		
		MCWBR	12.5	12.7	13.5	14.2	14.0	13.7	13.0	13.2	14.2	14.3	12.9	12.2		
(40) Clear-Sky Solar Irradiance	taub		0.388	0.437	0.508	0.541	0.549	0.547	0.534	0.489	0.458	0.478	0.443	0.392		
		taud	2.094	1.972	1.824	1.794	1.795	1.788	1.821	1.906	1.952	1.892	1.965	2.083		
	Ebn at Noon		768	773	753	755	757	757	763	789	790	718	693	729		
		Edh at Noon	121	155	197	214	217	218	210	189	171	165	138	116		
(44) All-Sky Solar Radiation	RadAvg		2.31	3.17	4.30	5.66	7.15	8.13	7.99	7.42	6.17	4.36	2.83	2.16		
	RadStd		0.20	0.38	0.37	0.56	0.26	0.22	0.14	0.17	0.14	0.23	0.26	0.22		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	+0.53	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)		N/A	N/A	N/A	+0.43	-0.39	N/A	N/A	N/A	N/A

Nomenclature: See separate page